

Introduction to Internet of Things

Lecture 10 (SDN)

Software Defined Networking (SDN)

- ▶ Software Defined Networking (SDN) is a networking approach where the control plane is separated from the data plane, and network control is managed centrally using software.
- ▶ SDN is a software platform (controller) that computes routes, enforces policies, and programs the network, replacing the autonomous routing decisions of traditional routers.
- ▶ SDN manages the network resources (switches, routers, links) from a centralized control plane.

Control Plane vs Data Plane

- ▶ The control plane is the part of network that determines how network packets are routed and processed, the brain. It carries the signaling and routing message traffic.
- ▶ The data plane is the part of network that forwards those packets to their destinations, the muscle. It carries the payload data traffic.
- ▶ Control plane builds routing tables and policies, whereas the data plan operates based on these rules to move traffic efficiently.

High Level

Control Plane

manages network routing protocols

OSPF, BGP, and MPLS protocols

Management Plane

manages the network's operations

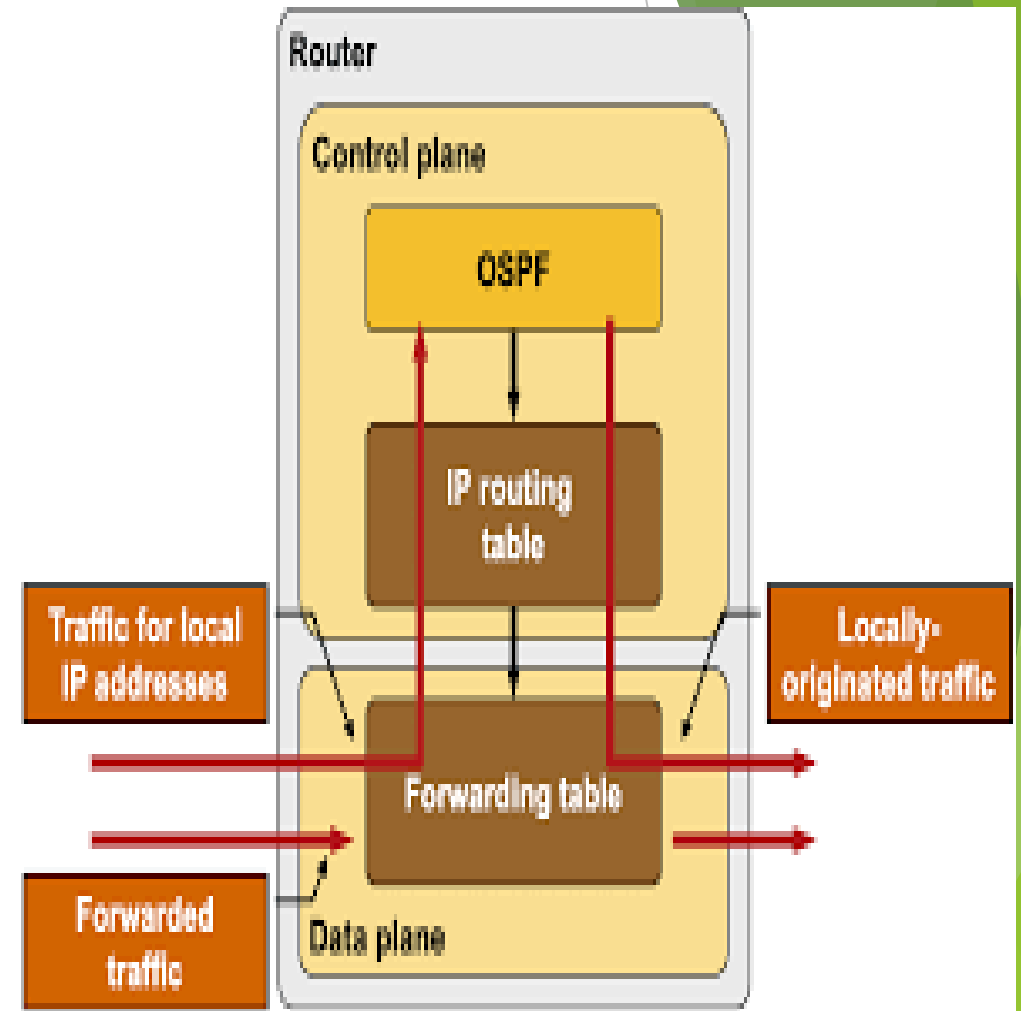
SNMP, SSH, and Telnet protocols

Low Level

Data Plane

forwards data packets between devices in the network

Ethernet, TCP/IP, and UDP

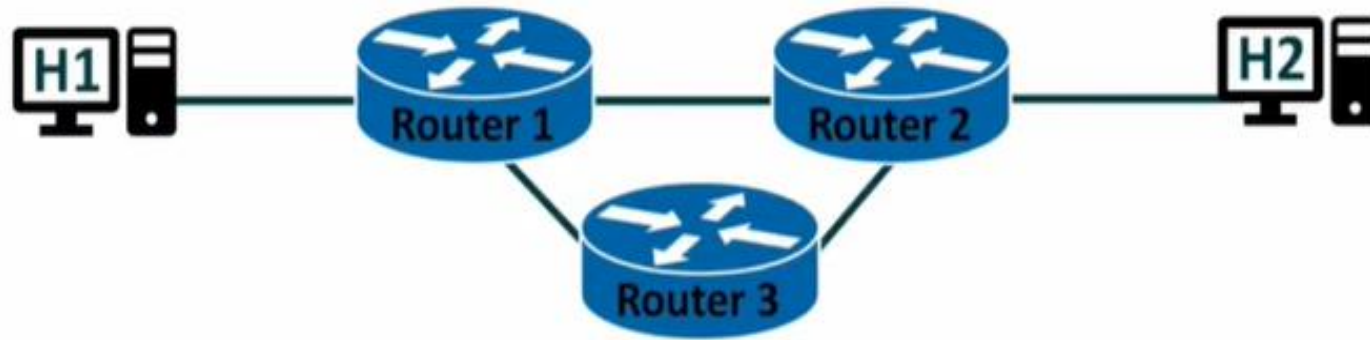


Router role

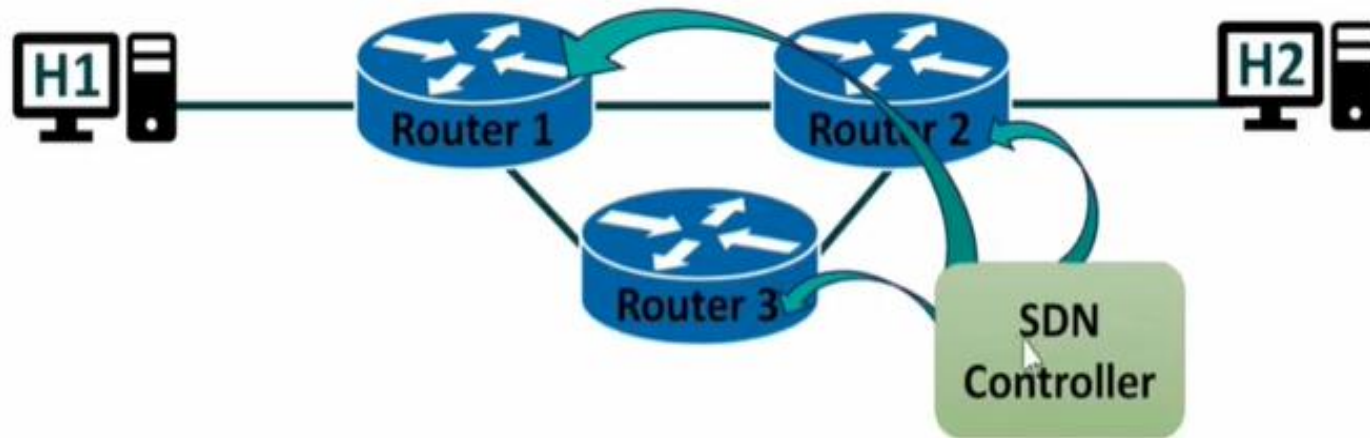
- ▶ The router process works as follows:
 1. Routing protocols build the routing table
 2. Best routes are selected
 3. Those best routes are copied into the forwarding table
 4. Packets are forwarded using the forwarding table

Traditional Networking vs SDN

- ▶ In traditional networks, routers will take the decisions of routing and forwarding the data.
- ▶ These decisions are made based on the routing algorithms and routing tables.
- ▶ In traditional networks, routers contain both:
 1. Control plane (routing decisions)
 2. Data plane (forwarding)
- ▶ Routing decisions are made using routing algorithms like:
 1. OSPF
 2. RIP
 3. BGP
- ▶ In SDN, routers will not take decisions, SDN controller will give the command to router where to forward the data. Software inside SDN controller will do decisions.

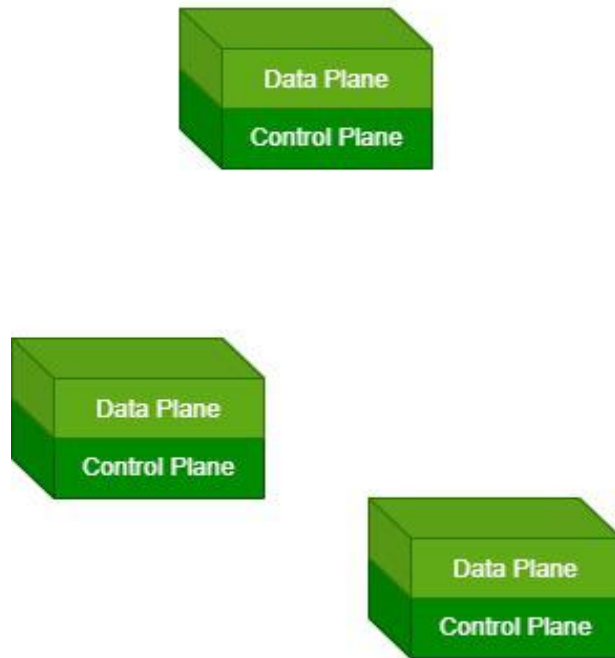


Traditional Networking

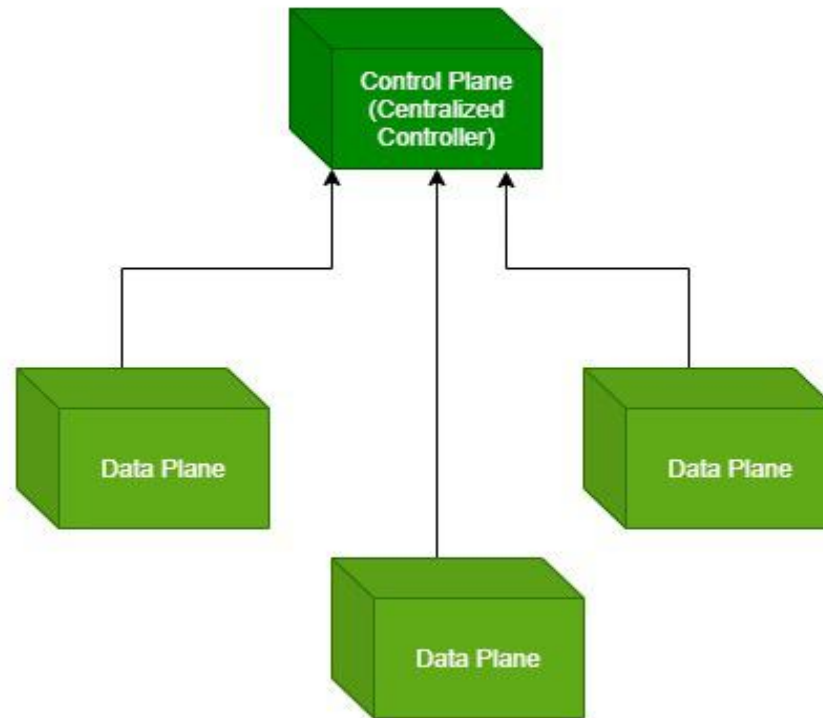


SDN – Software Defined Networking

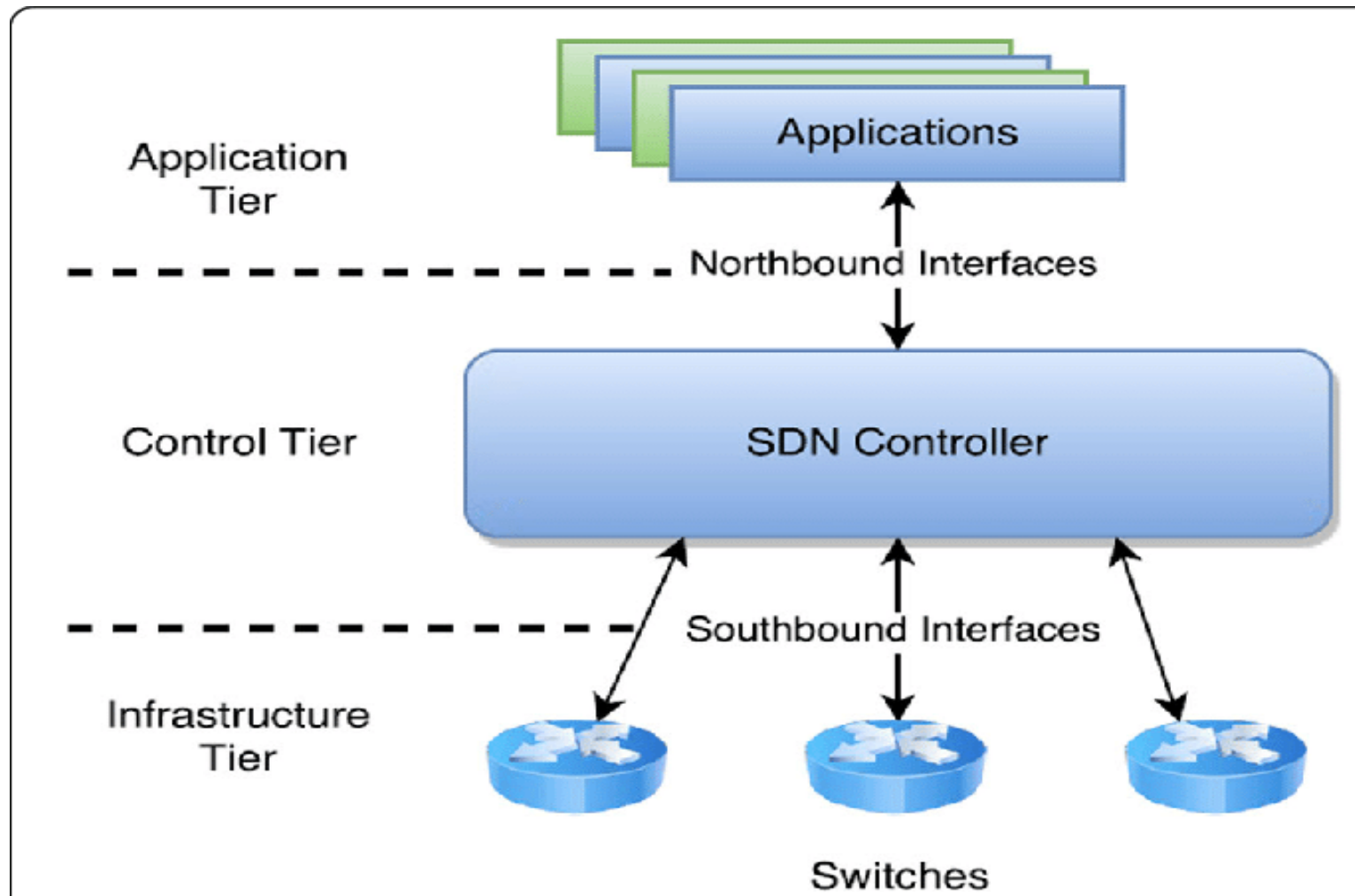
Traditional Network



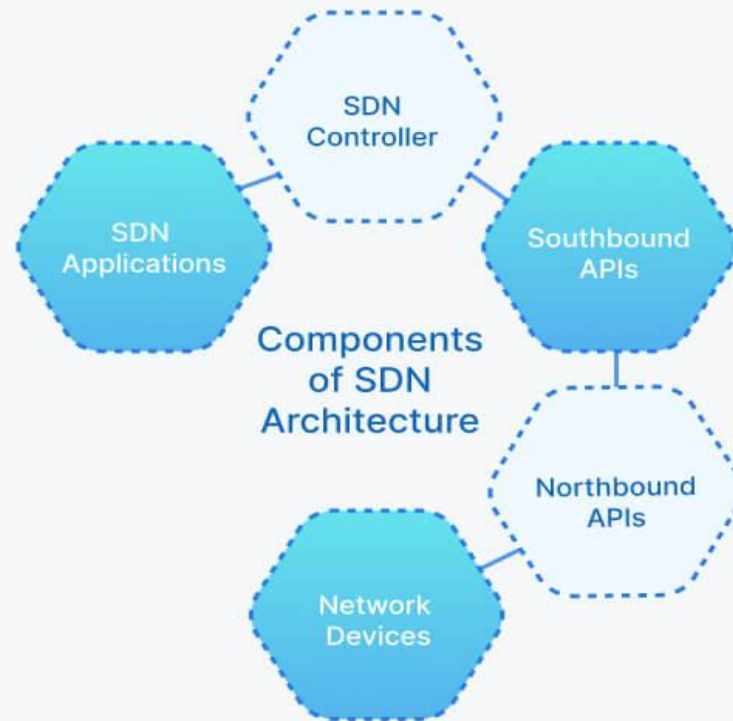
Software Defined Network



SDN Architecture



SDN Components



Factors	SDN	Traditional Networking
Architecture	Separates the control plane from the data place	Both the control plane and data planes are bound together
Nature	Software-centric	Hardware-centric
Control Plane	A centralized unit (SDN controller) handles the control plane	The control plane is distributed across all the networking devices
Configuration	Supports automatic configuration, centralized	Requires manual configuration, device-specific
Deployment Complexity	Reduces setup time and makes the network more adaptable	Time-consuming and less flexible
Programmability	Programmable	Not programmable
Scalability	High	Limited
Cost	Cost-effective, lower maintenance cost	Higher maintenance cost
Flexibility	Dynamic, adaptable	Static, siloed

Advantages of SDN

Benefits of software defined networking (SDN)

This slide talks about the benefits of software-defined networking, including control, efficiency, management, and visibility. The SDN enhances network connectivity for sales, customer support, private communications, and file sharing.

01

Control

- With the capacity to adjust a network's switch rules based on need, managers have better command over traffic flow
- Multi-tenant architecture in cloud computing relies heavily on this flexibility
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02

Efficiency

- Use of virtualized services minimizes the need for expensive hardware
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03

Management

- Without configuring peripherals, network managers may disseminate rules over switches using a centralized controller
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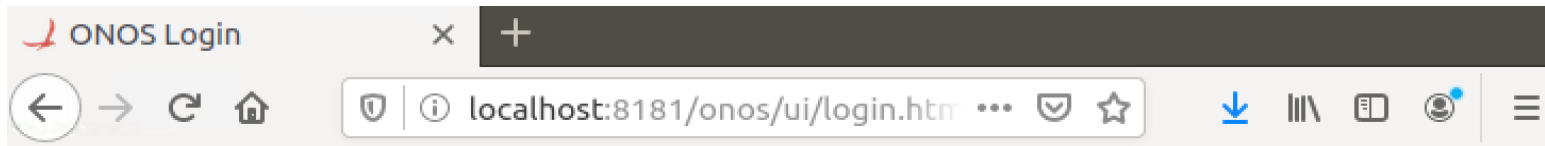
04

Visibility

- Centralized controller may recognize suspicious activity and redirect packets by monitoring traffic
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Use Cases	Benefits
Data Center Optimization	• Dynamic resource allocation • Automated provisioning and scaling • Traffic flow optimization
Cloud Networking	• CSPs can offer dynamic, on-demand cloud services • Enterprises can manage hybrid and multi-cloud environments • Proper bandwidth allocation and network slicing
Internet of Things (IoT) Networking	• Management of IoT devices with their connectivity needs • Prioritize critical data flows • Vibrant network changes



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IoT Perception Layer

